

PS500 DIN-Rail Switching Power Supply

SKU / Model Number: PS500-24V-20A

Manufacturer: PowerRail Electronics

Category: Electrical | ETIM Code: EC001405

PRODUCT OVERVIEW

The PS500 is a high-efficiency DIN-rail mounted switching power supply for industrial automation, PLCs, and field instrumentation. Features active PFC, wide input range, and comprehensive protections.

TECHNICAL SPECIFICATIONS

Output Voltage	24 V DC (adjustable 24-28 V)
Output Current	20 A continuous
Output Power	480 W
Efficiency	> 95% at full load
Ripple & Noise	< 100 mV p-p
Hold-up Time	> 20 ms at full load
Power Factor	> 0.95 (with PFC)

ELECTRICAL RATINGS

Input Voltage	100-240 V AC (85-264 V AC)
Input Frequency	47-63 Hz
Input Current	6.5 A at 120 V AC
Inrush Current	< 40 A
Leakage Current	< 1 mA

DIMENSIONS & MECHANICAL

Width	65 mm
Height	125 mm
Depth	127 mm
Mounting	35 mm DIN rail (TS35)
Weight	750 g

OPERATING CONDITIONS

Operating Temperature	-25 to 70 deg C
Derating	2.5% per deg C above 60 deg C
Storage Temperature	-40 to 85 deg C
Humidity	5 to 95% RH
Altitude	Up to 5000 m (with derating)

MATERIALS OF CONSTRUCTION

Housing	Aluminum extrusion, anodized
PCB	FR4, double-sided
Terminals	Nickel-plated copper
DIN Clip	Stainless Steel spring
Conformal Coating	Polyurethane (optional)

PERFORMANCE DATA / SELECTION TABLE

Load %	Voltage (V)	Ripple (mV)	Efficiency (%)
10%	24.1	< 30	88
25%	24.1	< 50	92
50%	24.0	< 70	94
75%	24.0	< 85	95
100%	24.0	< 100	95

CERTIFICATIONS & APPROVALS

IEC 62368-1 | CE | UL 508 Listed | CSA | ATEX Zone 2 | GL/DNV Marine

Note: Specific certification documents and test reports are available upon request. All products are manufactured under ISO 9001:2015 quality management system.

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